

KENYA NATIONAL EXAMINATIONS COUNCIL

KCPE 2003

SCIENCE

READ THESE INSTRUCTIONS CAREFULLY

Time:1 hour 40 minutes

1. You have been given this question booklet and a separate answer sheet. The question booklet contains 50 question
2. Do any necessary rough work in this booklet.
3. When you have chosen your answer, mark it on the **ANSWER SHEET**, not in this question booklet.
4. Use only an ordinary pencil.
5. Make sure that you have written on the answer sheet:

YOUR INDEX NUMBER

YOUR NAME

NAME OF YOUR SCHOOL

6. By drawing a dark line inside the correct numbered boxes mark your full Index Number(i.e School code number and the three-figured Candidate's number)in the grid near the top of the answer sheet.
7. Do not make any marks outside the boxes.
8. Keep the sheet as clean as possible and do not fold it.
9. For each of the question 1-50 four answers are given. The answers are lettered A,B,C and D. In each case only ONE of the four answers is correct. Choose the correct answer.
10. On the answer sheet the correct answer is to be shown by drawing a dark line inside the box in which the letter have chosen is written.

Example

In the Question booklet.

4. Which one of the following components of blood is involved in the clotting after injury?
 - A. Plasma
 - B. White blood cells
 - C. Red blood cells

D. Platelets

The correct answer is D

1 [A] [B] [C] [D] 14. [A] [B] [C] [D] 24. [A] [B] [C] [D] 34. [A] [B] [C] [D]

11. Your dark line MUST be within the box.

12. For each question ONLY ONE box is to be marked in each set of four boxes.

This question paper consists of 7 printed pages and 1 black page.

The Kenya National Examination Council, 2003.

1. The splitting of white light into its constituent colours is called.
 - A. Reflection
 - B. Refraction
 - C. Diffraction
 - D. Dispersion
2. A plastic bottle was placed in a carton and insulated materials put around the bottle as shown below. The bottle was filled with hot water
Which one of the following could help keep the water for longer time
 - A. Putting a lining of aluminum foil around the carton
 - B. Using insulated material with large air spaces
 - C. Using black insulated material
 - D. Using a metal box instead of carton
3. The diagram below represents a set up that can be used to demonstrate a certain effects of electricity

The effects demonstrated is made use of in making

- A. Electromagnetic
 - B. Switches
 - C. Light bulbs
 - D. Conductor
4. The function of hemoglobin in the blood is to
 - A. Fight germs
 - B. Transport food substance
 - C. Cause clotting of blood
 - D. Combine with oxygen
5. Which one of the following pairs of plants has root nodules
 - A. Tomato and beans
 - B. Cabbage and millet
 - C. Groundnuts and peas
 - D. Maize and potato
6. Bile mixes with the food in the
 - A. Stomach
 - B. Small intestine
 - C. Duodenum
 - D. Large intestine
7. Which one of the following is likely to happen if disease gradually kills antelopes in grassland inhabited by lions and vultures?

	Grass	lion	vultures
A.	Increase	decrease	increase
B.	Decrease	increase	increase
C.	Increase	increase	increase
D.	Increase	decrease	decrease

8. Which one of the following plants has roots similar to the type found in the carrots?
 - A. Onion
 - B. Cassava
 - C. Sweet potatoes
 - D. Arrow roots
9. The table below shows characteristic of some invertebrates
10. The mass of 1cm^3 of a substance is its
 - A. Volume
 - B. Density
 - C. Weight
 - D. Surface area
11. In which one of the following methods of separation would only one substance be removed from a mixture of two substances
 - A. Evaporation
 - B. Filtration
 - C. Decanting
 - D. Distillation
12. Which one of the following materials can be recycled?
 - A. Ash
 - B. Cow dung
 - C. Coffee husks
 - D. Scrap metals
13. In which of the following pair of levers is arrangement of positions load, fulcrum and effort the same
 - A. Crowbar and wheel barrow
 - B. Pair of scissors and nail cutter
 - C. Fishing rod and beam balance
 - D. Nutcracker and pincers
14. Which one of the following weather instruments does not work on the fact that matter occupies space?
 - A. Rain gauge
 - B. Air thermometer
 - C. Wind sock
 - D. Wind vane
15. The soil that can be used to make long ribbon has
 - A. Cracks when dry
 - B. Large air space
 - C. Rough texture
 - D. Low capillarity
16. The diagram below represents a circuit

Which one of the following statements is TRUE about the circuit? The

- A. Bulb uses power from three cells
- B. Bulb uses power from one cell
- C. Cells are connected in series
- D. Cells are connected in parallel

17. Which one of the following pair consists of renewable sources of energy

- A. Firewood and kerosene
- B. Water fall and petrol
- C. Biogas and charcoal
- D. Wind and coal

18. The diagram below represents a set up used by pupils to compare the rate at which heat is conducted in metals

Pupils would make wrong conclusion if they used?

- A. Metals of different thickness
- B. Different type of pins
- C. Metals of different length
- D. Different types of metals

19. The diagram below represents a straw fitted in an airtight bottle containing a liquid

The sucking liquid is difficult because

- A. The straw does not reach the bottle
- B. The liquid does not fill the bottle
- C. The air inside the straw prevents the liquid from rising
- D. Air pressure does not act on the liquid

20. In an experiment equal amount of the following pairs of liquids were put in four identical containers:

- i) Kerosene and pure water
- ii) Salty and kerosene
- iii) Pure water and engine oil
- iv) Methylated spirit and engine oil

A fresh egg was then placed in each container would the egg NOT sink to the bottom?

- A. (i)
- B. (ii)
- C. (iii)
- D. (iv)

21. Choose a group of substance where all the three are acidic

- A. Toothpaste, clovers, wood ash
- B. Wood ash, sour milk, vinegar
- C. Lemon juice, toothpaste, baking powder
- D. Clover, vinegar, sour milk

22. Which one of the following represents the correct order of stages in the life of a butterfly?
- A. Adult → egg → larva → pupa
 - B. Egg → pupa → larva → adult
 - C. Egg → larva → pupa → adult
 - D. Adult → egg → pupa → larva
23. Which one of the following plants has separate male and female flowers?
- A. Sorghum
 - B. Maize
 - C. Oranges
 - D. Sisal
24. Steam can be used to drive a wheel as in the set up shown in the diagram below
- Which one of the following cannot increase the speed of the rotating wheel?
- A. Using blades made from lighter materials
 - B. Increase the size of the blade
 - C. Increase the size of the hole
 - D. Increase the amount of heat
25. Which one of the following is correct statements about pulleys? Pulleys
- A. Increase friction
 - B. Reduce the weight of the load
 - C. Change direction of force
 - D. Increase speed
26. Which one of the following is an example of force?
- A. Weight
 - B. Energy
 - C. Pressure
 - D. Work
27. Which one of the following is not an example of an inclined plane?
- A. Crowbar
 - B. Staircase
 - C. Wedge
 - D. Screw
28. Which one of the following will not change when loam soil is added to sandy soil
- A. Texture
 - B. Capillarity
 - C. Colour
 - D. Fertility
29. Choose a pair where any of the methods can be used for making water safe for drinking
- A. Addition of chemical, filtration
 - B. Decantation and boiling
 - C. Boiling, addition, decantation
 - D. Filtration, decantation

30. The chart below represents a simplified classification of plants.

Which plants are represented by X and Y?	
X	Y
A. Cyprus	mushroom
B. Sugarcane	toadstool
C. Moss	ferns
D. Grass	bread mould

31. The tips of threads-like structures found in the maize cob are

- A. Anthers
- B. Filament
- C. Styles
- D. Stigma

32. Which one of the following helps a cactus plants reduce loss of water

- A. Deep root
- B. Swollen stem
- C. Thorn like leaves
- D. Green stem

33. Which one of the following pairs of insects has a larva stage in their life cycle

- A. Moth and bee
- B. Housefly and cockroach
- C. Mosquito and grasshopper
- D. Termite and locust

34. During the growth of tadpole, the following parts develop in stages

- I) Hind limbs
- II) Lungs
- III) Front limbs
- IV) External gill
- V) Tail

The correct order in which the stage occurs is

- A. (iv) (i) (v) (ii) (iii)
- B. (iii) (i) (ii) (iv) (v)
- C. (v) (iv) (ii) (i) (iii)
- D. (ii) (iii) (iv) (i) (v)

35. A plastic cup was inverted over water in a basin as shown in the diagram below. It was observed that the water did not enter the cup

Which of the following could be done to make water enter the cup?

- A. Pushing the cup to the bottom
- B. Adding more water to the basin
- C. Reducing the size of the cup by cutting

- D. Making a hole at the bottom of the cup
36. Which one of the following activities will correct conclusion be drawn about evaporation of equal amounts of water and spirits?
- A. Putting the liquid in the bottle tops and placing both in the sun
 - B. Putting one liquid in the bottle top and the other shoe polish tin then placing the two in the sun
 - C. Putting the liquid in the bottle tops then placing one in the sun and heating the other with a candle
 - D. Putting one liquid in the bottle top and the other in a shoe polish tin lid then heating both of them with a candle
37. Which one of the following is not true about pressure in liquid increase when the
- A. Amount of liquid is increased
 - B. Diameter of container is reduced
 - C. Depth of the liquid is increased
 - D. Container with liquid is raised
38. Which one of the following is difference between solids and gases?
- A. Solids has mass whereas gases do not
 - B. Solid have definite shape whereas gases do not
 - C. Solids have definite shape whereas gases take the shape of the container
 - D. Solids can change to liquid whereas gases cannot change to liquid
39. When lime powder was mixed with juice obtained from hibiscus flower, it changed colour. Which of the following substance would not make hibiscus flower juice change to the same colour
- A. Wet wood ash
 - B. Sour milk
 - C. Baking powder
 - D. Chalk dust
40. The diagram below represents a lever in use

The load would be moved with least effort if distance is increased between points

- A. J and M
 - B. K and L
 - C. K and M
 - D. J and K
41. Which one of the following pairs of planets consists of the smallest planets and the largest in the solar system
- A. Earth and mars
 - B. Mercury and Jupiter
 - C. Venus and Uranus
 - D. Pluto and Saturn
42. The chart below represents weather records for four days
- On which day did it rain immediately after being windy

- A. Monday
 - B. Tuesday
 - C. Wednesday
 - D. Thursday
43. Which one of the following pairs of forms of energy can be transmitted only when there is a medium
- A. Heat and light
 - B. Sound and electricity
 - C. Sound and light
 - D. Heat and electricity
44. The following are function of stem
- i) Support the shoot
 - ii) Storage of food
 - iii) Transport water to the leaves
 - iv) Manufacture food for plants
- Which function are performed by all stems?
- A. (i) and (iii)
 - B. (iii) and (iv)
 - C. (i) and (ii)
 - D. (ii) and (iv)
45. The diagram below represents a pulley in use
- Which one of the following statements is true about the distance moved by the load and the effort?
- A. Three times the distance moved by the effort
 - B. Twice the distance moved by the effort
 - C. The same distance as that moved by effort
 - D. Half the distance moved by the effort
46. Which one of the following pairs of drug consists of stimulant and depressant
- A. Khat and tobacco
 - B. Cocaine and alcohol
 - C. Cobbles and glue
 - D. Bhang and caffeine
47. Which one of the following substance is prohibited in Kenya when used as drug?
- A. Beer
 - B. Tobacco
 - C. Khat
 - D. Cobbler's glue
48. The activities below are carried out when separating a mixture of salt, iron fillings and sand but not in their correct order
- i) Filter the mixture through the sieve

- ii) Add water to the mixture
- iii) Leave mixtures in the sun
- iv) Pass a magnet through the mixture

The correct order of activities is

- A. (iv) (ii) (i) (iii)
- B. (ii) (iii) (i) (iv)
- C. (ii) (iii) (iv) (i)
- D. (iv) (i) (ii) (iii)

49. When transparent bottle filled them with garden soil and water shaken and left to stand, layers formed

The order in which the layers formed depend on

- A. Density of components
- B. Amount of water
- C. Size of components
- D. Mass of the components

50. Which one of the following statements is not true about bhang and khat? Both

- A. Are addictive drugs
- B. Are from plants materials
- C. Are from prohibited drugs
- D. Affects the nervous system